Question One: If $P(B) = 0.25$ and $P(A - B) = 0.25$ then we get $P(A|\Omega) - P(B)P(A|B) =$

(A) 0.25 B) 0.5 C) 0 D) 0.2 E) None

Question Two: For some disjoint and independent events A, B with $P(A) = 0.1$ we have $P(B) =$

A) 1 B) 0.3 (C) 0 D) 0.4 E) None

Question Three: Let X be a discrete random variable with $E(X) = \bar{X}$. If $P(X = \bar{X}) = 0.8$, then $P(X \neq \bar{X}) =$

A) 0 B) 0.8 C) 0.75 (D) 0.2 E) None

Question Four: For some independent events A, B with $P(A)P(B) = 0.2$ then we have $P(\bar{A} \cup \bar{B}) =$

A) 0.9 (B) 0.8 C) 0.7 D) 0.6 E) None

Question Five: Let A, B be two events with equal probability and $P(\bar{A} \cap B) + P(\bar{B} \cap A) = 1.6$, then $P(A) =$

A) 0.2 B) 0.1 C) 0.8 D) 0.3 (E) None

Question Six: Let X be a discrete random variable with $R_X = \{0, 1, -1\}$ then we have $E(X^2) + E(X^4) - 2E(X^6) =$

(A) 0 B) 1 C) 2 D) -1 E) None

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Question Seven: Let f be a continuous strictly increasing function on $\mathbb{R}$. Then we have $\int_{-\infty}^{\infty} f(x)(\delta(x+1)-\delta(x-1))\,dx$ is

A) zero (B) negative C) positive D) equal to one E) None

Question Eight: For the joint PDF $f(x,y) = 2x^{b}$ when $0<x<1$, $0<y<1$ and zero otherwise, the positive constant b equals

A) 2 B) 0.5 C) 0 (D) 1 E) None

Question Nine: For the joint PDF $f(x,y) = 0.5$ when $0<x<1$, $0<y<2$ and zero otherwise, we have $P(0<Y<0.2) =$

(A) 0.1 B) 0 C) 0.25 D) 0.5 E) None

Question Ten: A discrete random variable X with PDF

$P(X=x) = a(\ \delta(x)+\delta(x-1)+\delta(x+1)+\delta(x-2)\ )$; where $a>0$ is a constant, then we have $a + P(X=0) + P(X=1) + P(X=-1) =$

A) 0 (B) 1 C) 0.75 D) 0.5 E) None

Question Eleven (✓, (X)): For continuous random variables X, Y we always have $Cov(-X-1,-Y) \geq -\sigma_X\sigma_Y$ X

Question Twelve (✓, (X)): For continuous random variable X with CDF $F(x) = 1-e^{-\sqrt{x}}$ when $x>0$ and zero otherwise; then the PDF is $f(x) = e^{-\sqrt{x}}/\sqrt{x}$ X

Question Thirteen ((✓) X): If $x \in (-1,1)$; then $u(-x-1)+\delta(x-2)=1$ X

Question Fourteen (✓ (X)) For some random variable X , if the CDF $F(x) = x^3, x \in (0,1)$, then $P(X \in (0,0.1)) + P(X=0.5) = 0.01$ X

Question Fifteen (✓, (X)): Given the CDF

$$F(x) = \begin{cases} 0 & x<1 \\ 0.25 & 1 \leq x<2 \\ 1 & 2 \leq x \end{cases}$$ ✓

We have $P(X=2) = 3/4$

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Q1: $P(A|\Omega) = P(A)$

$P(A) - P(B)\,\dfrac{P(A\cap B)}{P(B)} = P(A) - P(A\cap B) = P(A-B)$

Q2: $P(A\cap B) =$ zero (disjoint event)

$0.1\,P(B) = 0 \longrightarrow P(B) = 0$

Q3: $P(X \neq \bar{X}) = P(\bar{\bar{X}}) = 1 - P(\bar{X}) = 0.2$

Q4: $P(\bar{A}\cup\bar{B}) = P(\overline{A\cap B}) = 1 - P(A\cap B)$

$= 1 - P(A)\cdot P(B) = 1 - 0.2 = 0.8$

Q5: السؤال حذف

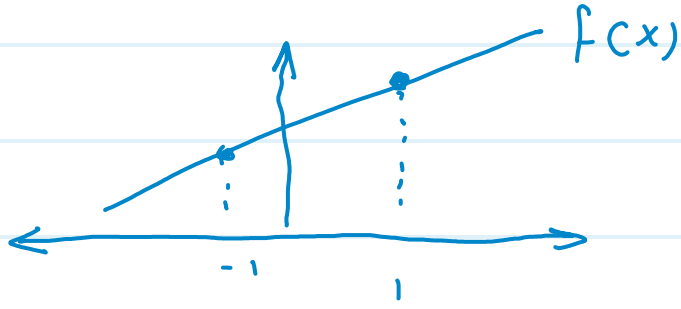
Q6: $EX^2 = 0\,p(0) + 1\,p(1) - 1\,p(-1) = EX^4 = EX^6$

$EX^2 + EX^4 - 2EX^6 = 2EX^2 - 2EX^2 = 0$

Q7: $\int_{-\infty}^{\infty} f(x)\,\delta(x-(-1))\,dx - \int_{-\infty}^{\infty} f(x)\,\delta(x-1)\,dx$

$= f(-1) - f(1) = -$

$f(-1) < f(1)$



Q8: $\iint_{R_{XY}} f(x,y)\,dx\,dy = 1$

$$\int_0^1 2x^b \left(\int_0^1 dy\right) dx = \frac{2x^{b+1}}{b+1}\Big]_0^1$$

$$\frac{2}{b+1} = 1 \longrightarrow b+1 = 2 \longrightarrow b = 1$$

Q9: $f_Y(y) = \int_{R_X} f(x,y)\,dx = \int_0^1 \frac{1}{2}\,dx = \frac{1}{2}$

$$P(0<Y<0.2) = \int_0^{0.2} \frac{1}{2}\,dy = \frac{1}{2}(0.2-0) = 0.1$$

Q10: $P(X=x) = P_r(x_1)\,\delta(x-x_1) + \dots + P_r(x_n)\,\delta(x-x_n)$

X	P(x)
0	a
1	a
-1	a
2	a

$\sum_{x \in R_X} p(x) = 1$

$4a = 1 \rightarrow a = 1/4$

Q11: $Corr(X,Y) = \frac{Cov(X,Y)}{\sigma_X \sigma_Y}$,

$$Corr(-X-1,-Y) = +Corr(X,Y) = \frac{Cov(-X-1,-Y)}{\sigma_X \sigma_Y}$$

$Cov(-X-1, -Y) \leq \sigma_X \sigma_Y$ ← عشان يكون ✗

$-Cov(-X-1, -Y) \geq -\sigma_X \sigma_Y$ محصور بين $[1, -1]$

Q12: $f(x) = \frac{dF(x)}{dx} = \frac{d}{dx}(1 - e^{-\sqrt{x}})$

$= -\frac{e^{-\sqrt{x}}}{-2\sqrt{x}} = \frac{e^{-\sqrt{x}}}{2\sqrt{x}}$ ✗

Q13: $u(c-x) = \begin{cases} 0, & x > c \\ 1, & x \leq c \end{cases}$

since $x \in (-1, 1)$ then $\delta(x-2) = 0$

so... $u(-1-x) + \delta(x-2) = 0$ ✗

Q14: $x \in (0, 1)$ so x is a continuous r.v

so $p(x = 0.5) = 0$ ✗

$p(x \in (0, 0.1)) = \int_0^{0.1} f_x(x)\,dx = F(0.1) - F(0) = 0.001$

Q15: $p(1) = 1/4$, $p(2) + p(1) = 1$

$p(2) = 1 - p(1) = 1 - 1/4 = 3/4$ ✓